

Multichamber J

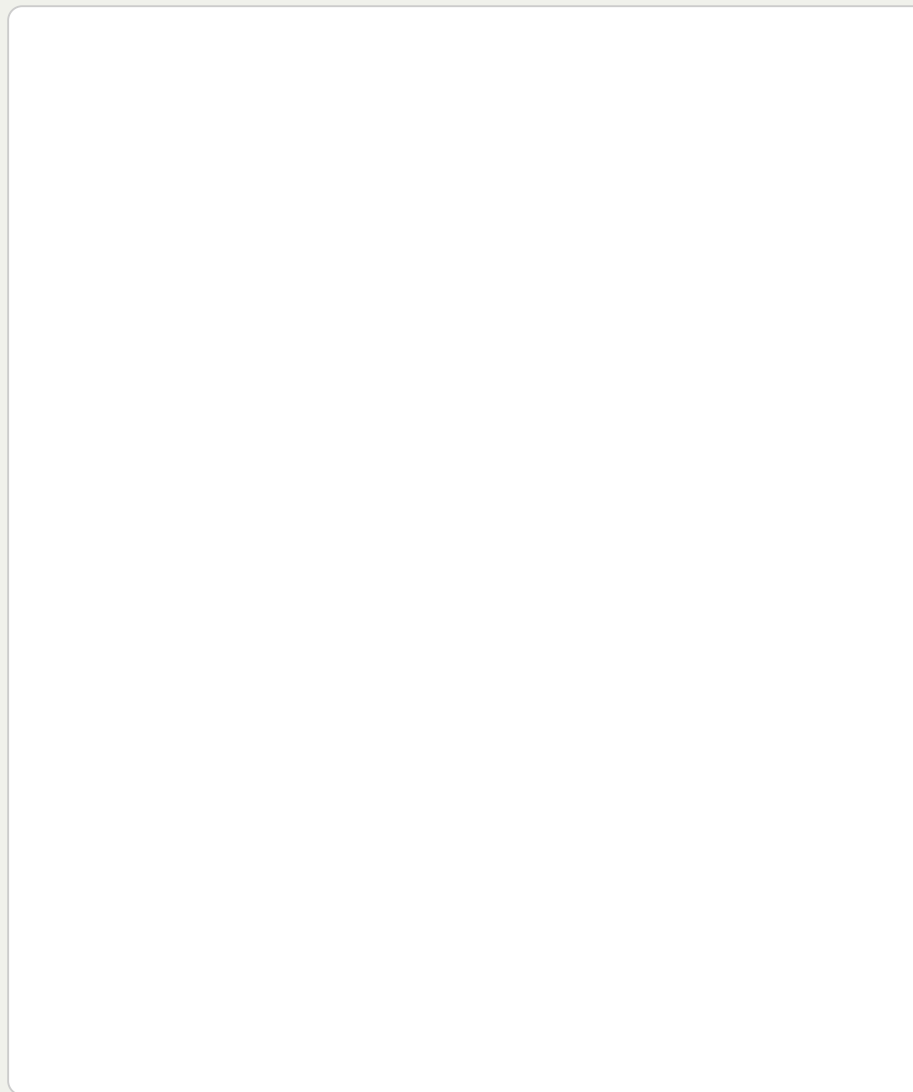
Michael Raba, MSc Candidate

Created: 2025-0

Muffler System

ate at University of Kentucky

9-15 Mon 12:38

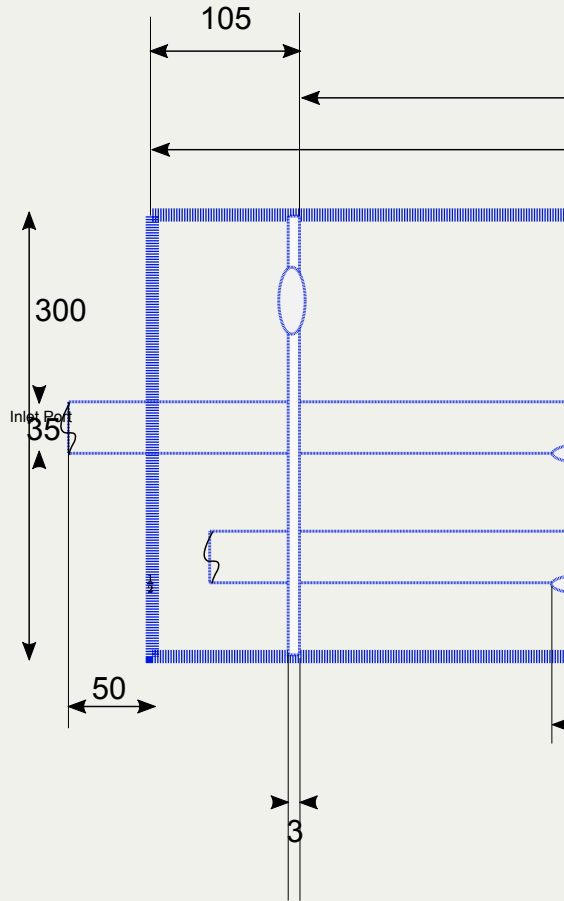


der Internal Geometry



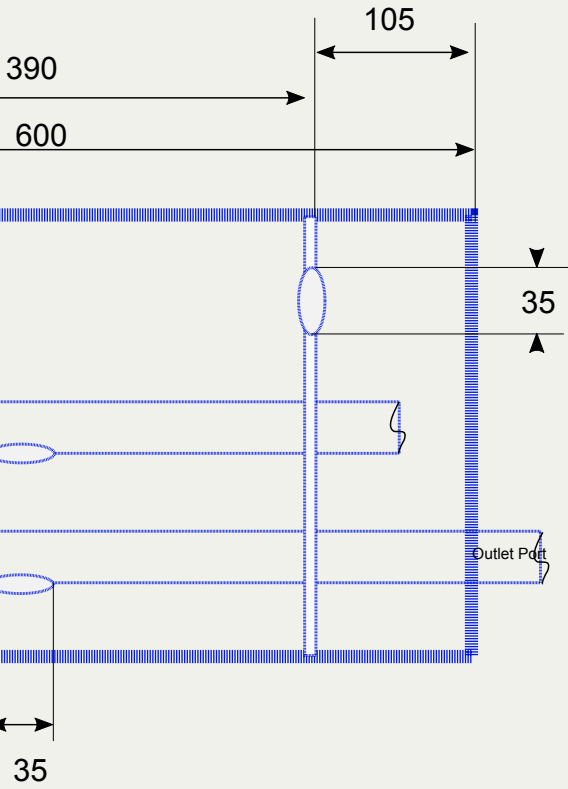
Dimension

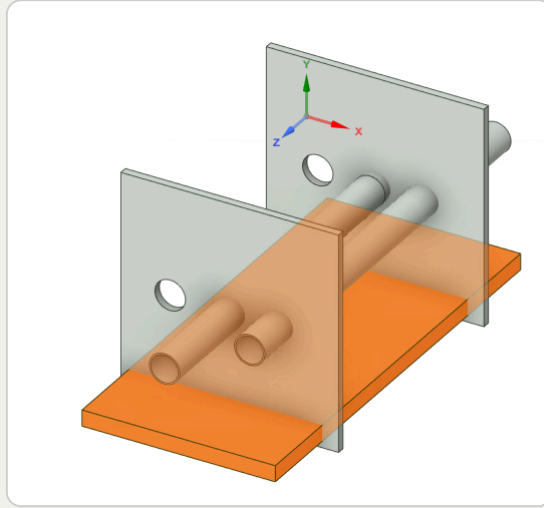
dimension



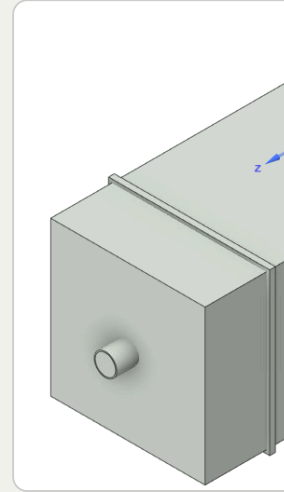
nsions

al units in mm



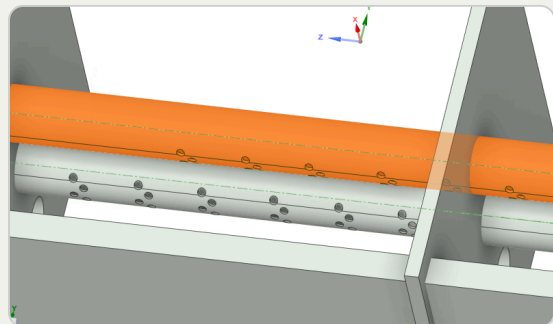


Part 1 — Chamber and Baffle

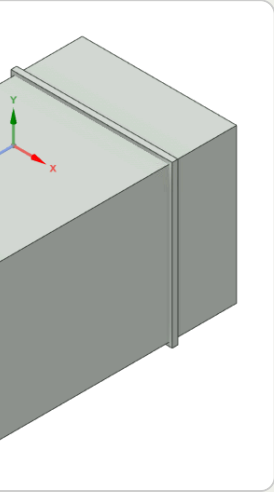


Part 2 — Front Panel

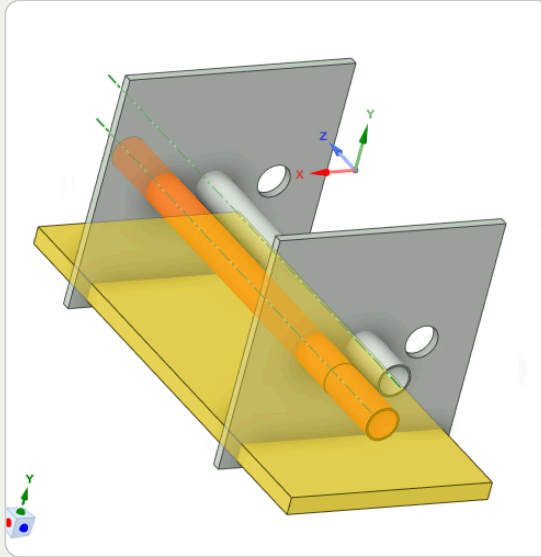
Part 4 — Showing perforates (aimed at fiberglass)



Muffler Subcomponents

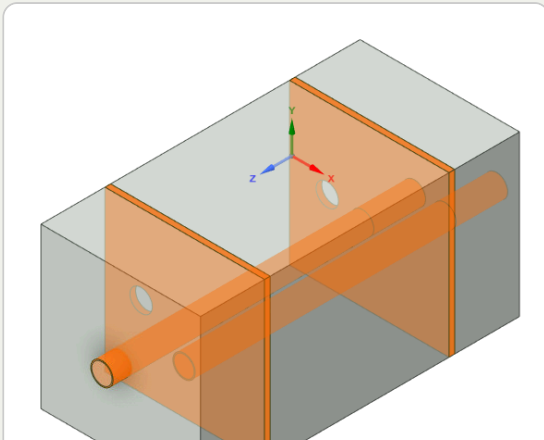


fluid domain



Part 3 — Fiberglass Absorbant (gold)

Part 5 — Final Assembly View



Simulated Transmission Loss (0–1000 Hz) by a

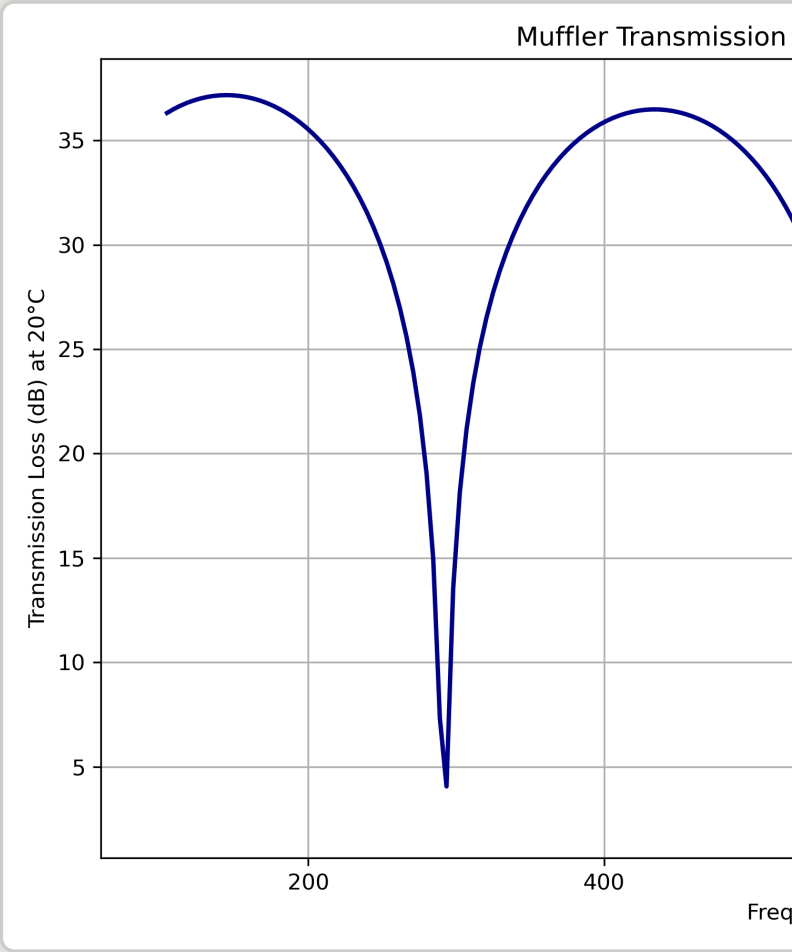
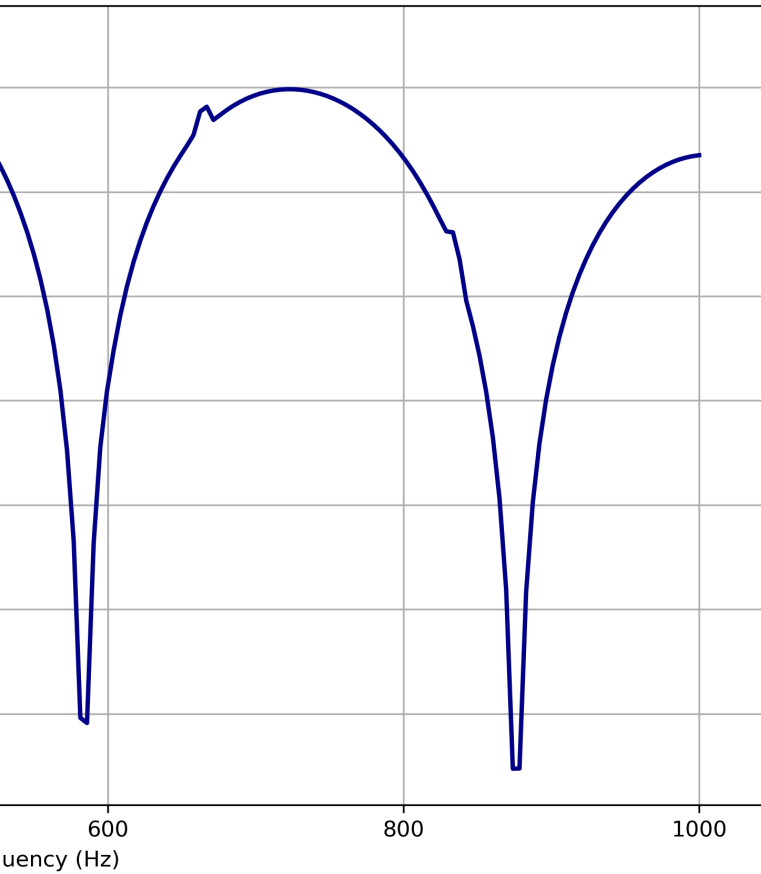


Figure: Transmission Loss curve of the m

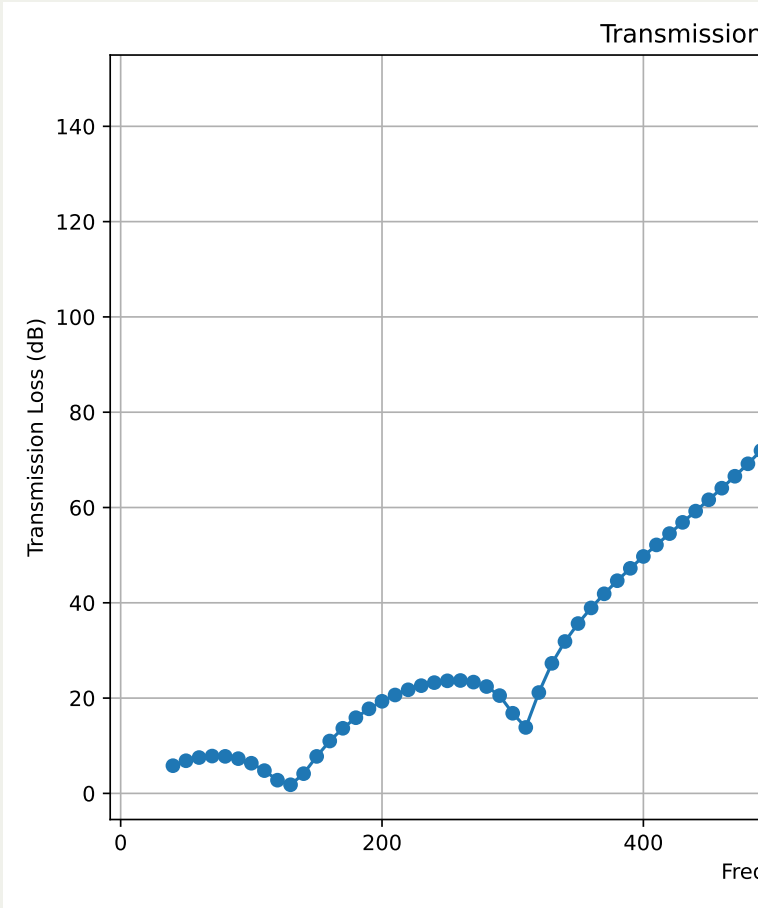
Simulation

approximating muffler walls as fluid at 20 deg C

Loss vs Frequency at 20°C



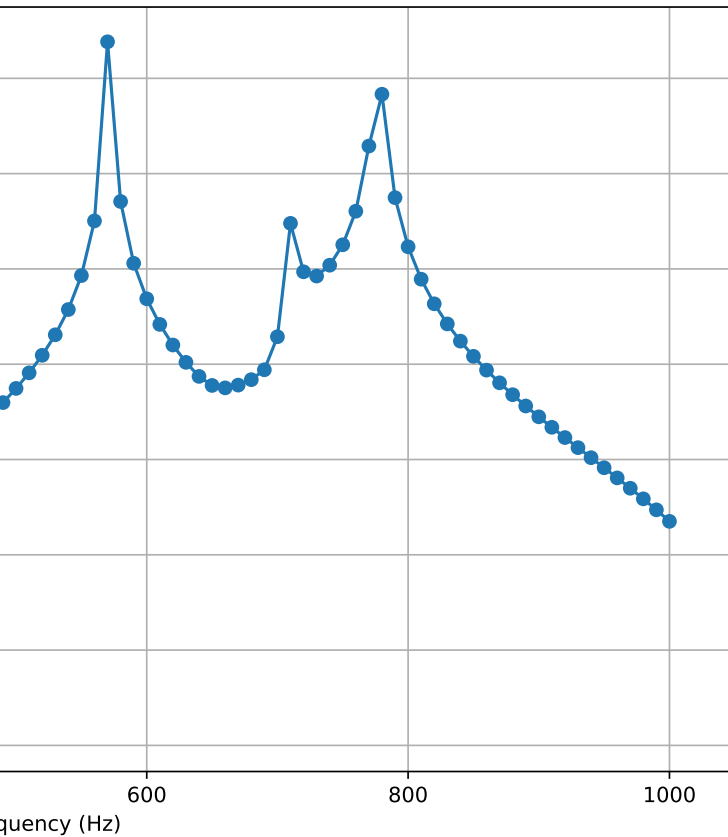
muffler between 5 Hz and 1000 Hz at 20°C.



Simulation

Loss (0–1000 Hz) Simlab model

Loss vs Frequency



Sidlab and Ansys Fi

SIDLAB Model

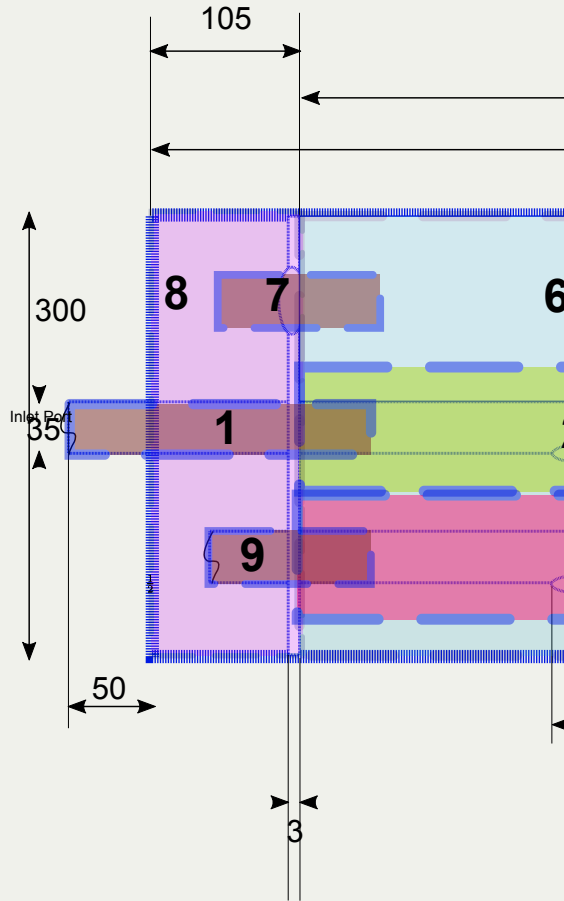
- **File:** Mark3Sid.zip
- **Created with:** SIDLAB 5.1
- [↓ Download SIDLAB File](#)

le Download Center

ANSYS Simulation

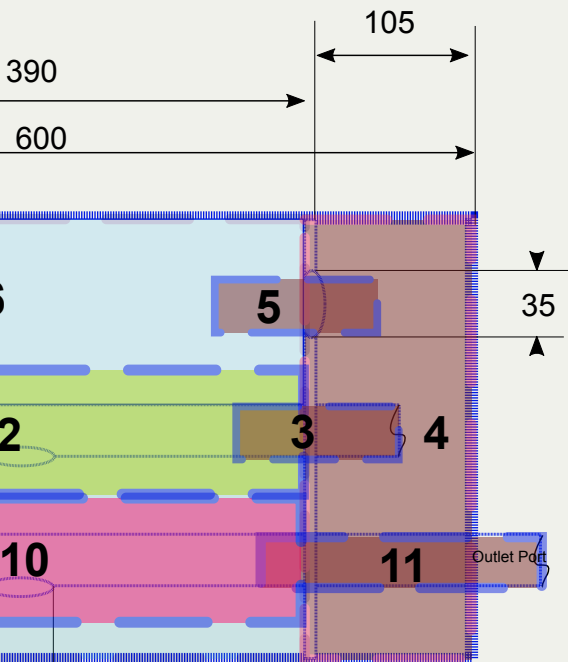
- **File:** Mark-I-MDF-cleared-data.wbpz
- **Created with:** ANSYS 2023 R2
- [↓ Download ANSYS File](#)

dimension



Components

al units in mm

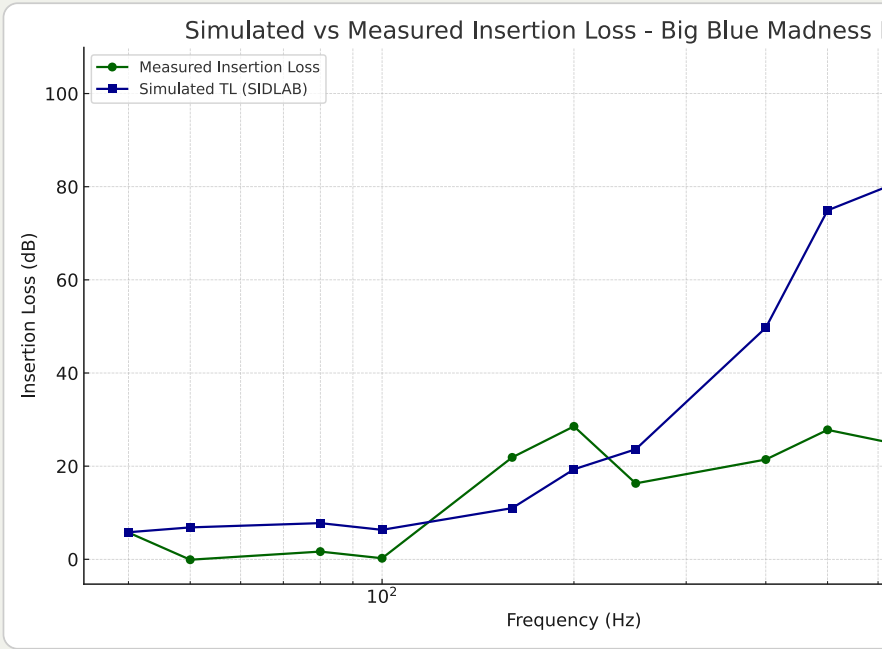


Sidlab Components

- | | |
|---------|----------|
| 1. Pipe | 7. Pipe |
| 2. QWT | 8. QWT |
| 3. Pipe | 9. Pipe |
| 4. QWT | 10. QWT |
| 5. Pipe | 11. Pipe |
| 6. QWT | |

Simulated vs Measured

Measured vs Simulated TL



Measured Insertion Loss



Insertion Loss Explanation

Insertion Loss (IL) quantifies how much sound is attenuated when a muffler is added to the system.

General formula:

$$IL = 10 \log_{10} \left(\frac{P_{\text{baseline}}}{P_{\text{muffler}}} \right)$$

Because our data is already in decibels (dB), this simplifies to:

$$IL = \text{Power}_{\text{baseline}} (\text{dB}) - \text{Power}_{\text{muffler}} (\text{dB})$$

1. Munjal ML. *Acoustics of Ducts and Mufflers*. 2nd ed. V
<https://doi.org/10.1002/9781118443125>
2. Dokumacı E. *Duct Acoustics: Fundamentals and Appli*
Press; 2021. ISBN: 9781108840750. <https://doi.org/10>

Note: These references are foundational texts in muffler and
schematic development, and

ences

Works

Wiley; 2014. ISBN: 9781118443125.

Applications to Mufflers and Silencers. Cambridge University
.1017/9781108840750

and duct acoustics and were consulted for system modeling,
and transmission loss analysis.